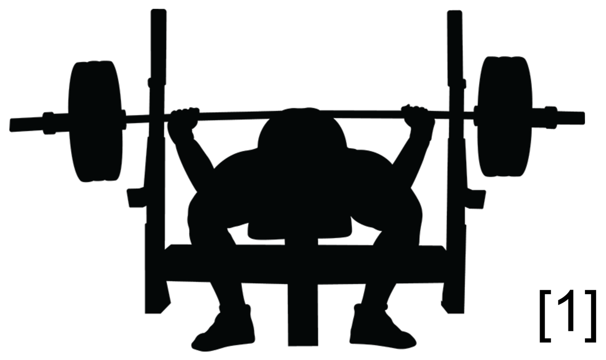

Bench Press: Grip Distance Effect on Muscle Activation

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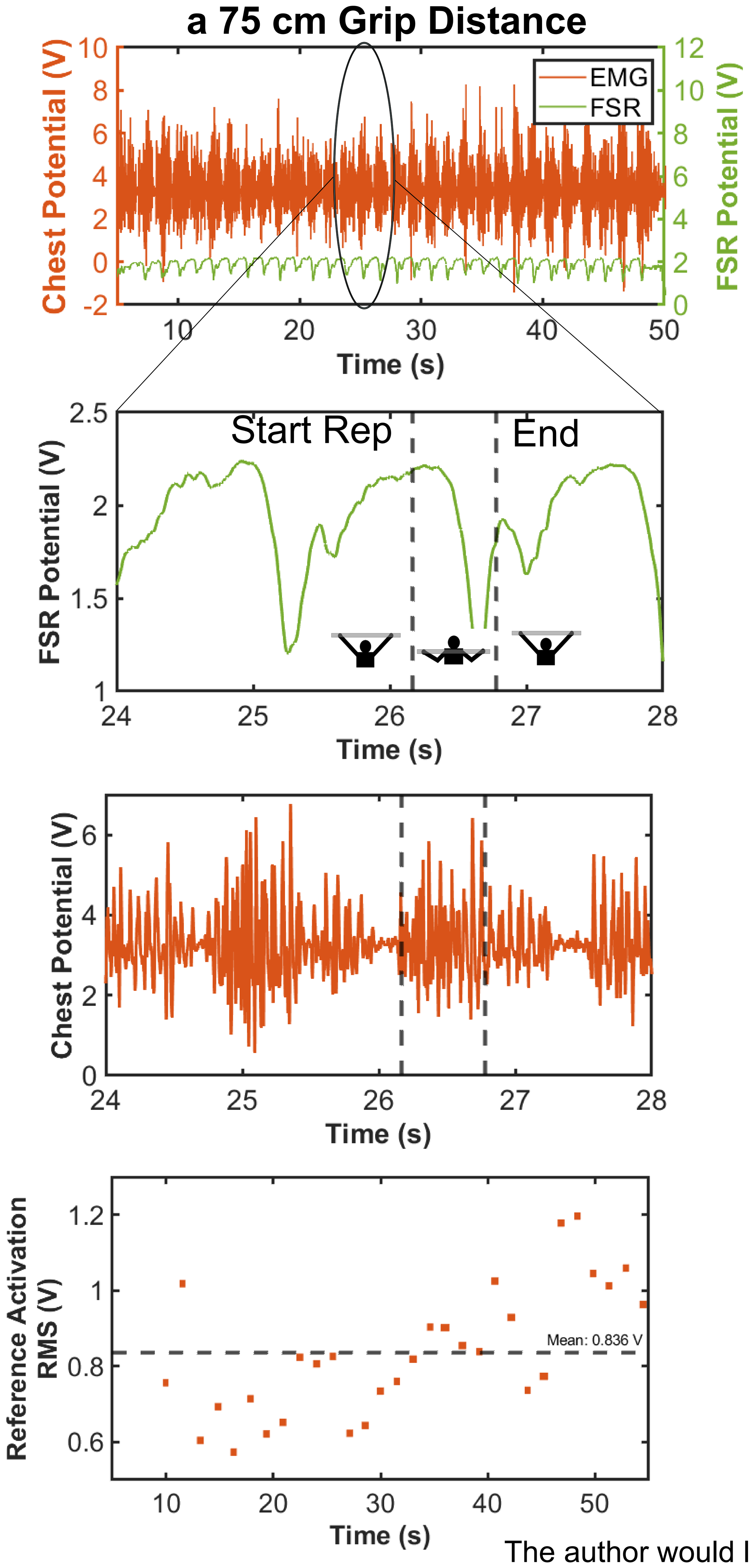
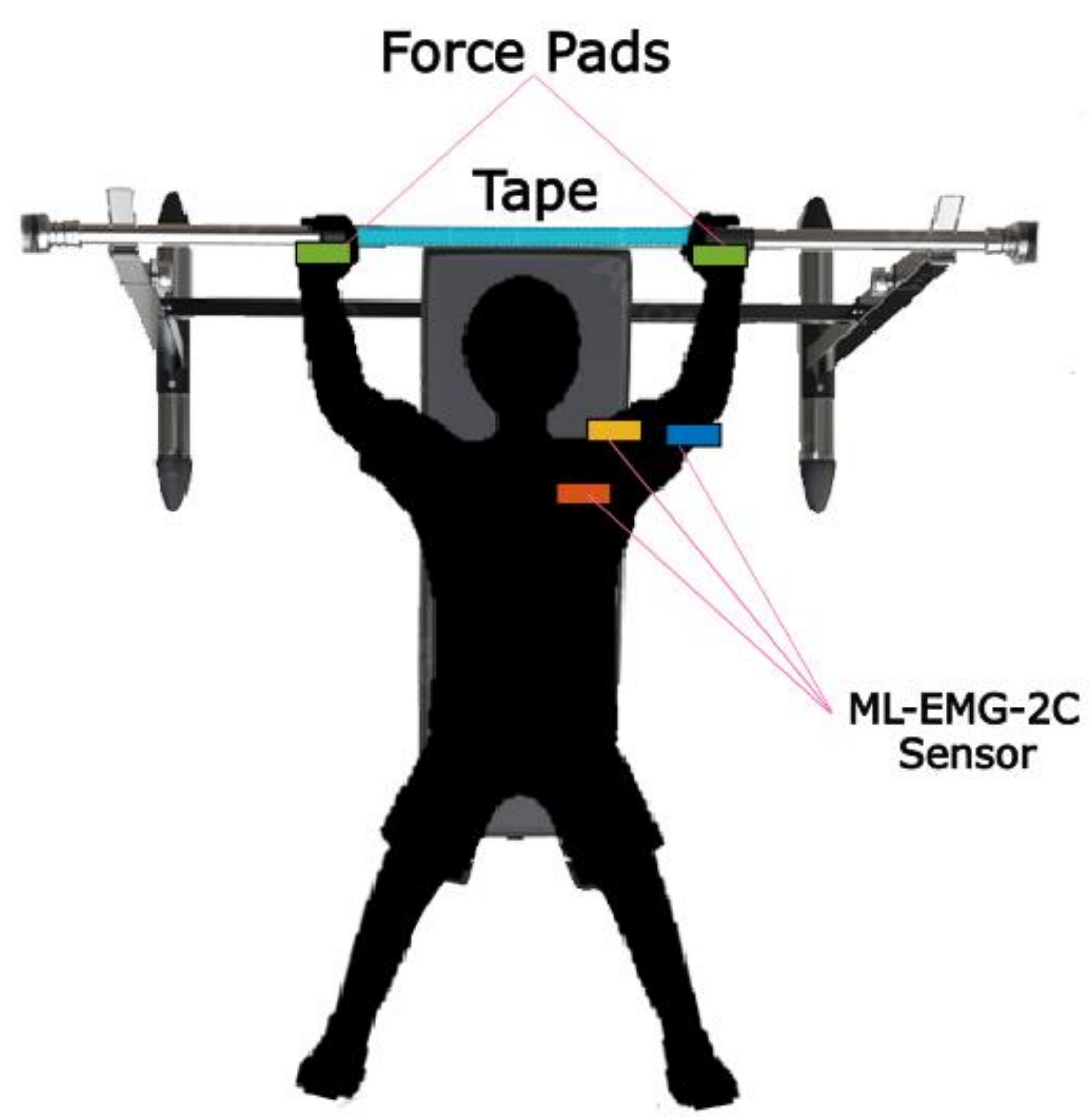
Abstract

Understanding how grip distance on the bench press affects muscle activation in the chest, shoulders, and triceps allows weightlifters to target specific muscle groups, as higher activation correlates with greater strength improvement. In this experiment, surface electromyography sensors were attached to these muscles to determine the activation, and an experienced lifter performed 30 repetitions with a 15 kg barbell across eight evenly spaced grip distances ranging from 15 cm to 85 cm. The activation profiles for each muscle groups are determined during repetitions at each grip distance. As grip distance increased from 15 to 85 cm, chest activation rose by 133%, while activation decreased by 25% in the shoulders and 73% in the triceps. Therefore, wider grips emphasize chest development, whereas narrower grips place greater demand on the shoulders and triceps.

Experimental Design

EMG and FRS signals at a 75 cm Grip Distance





$$\text{RMS} = \sqrt{\frac{\sum_i x_i^2}{n}}$$

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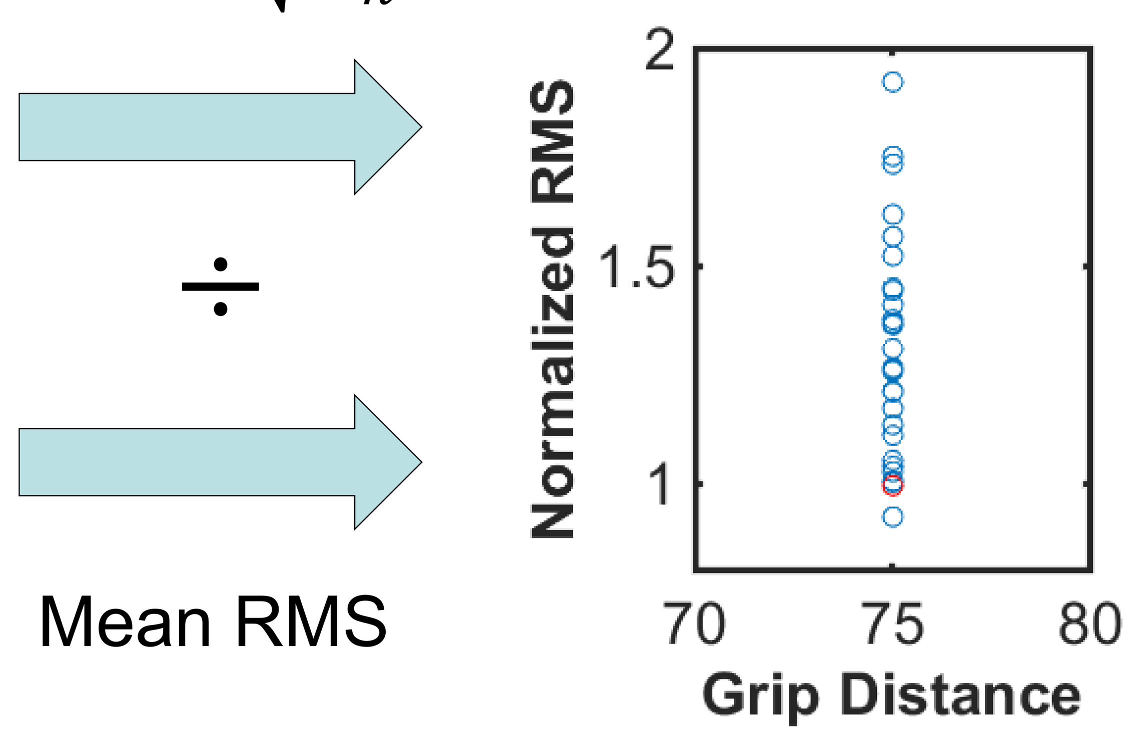
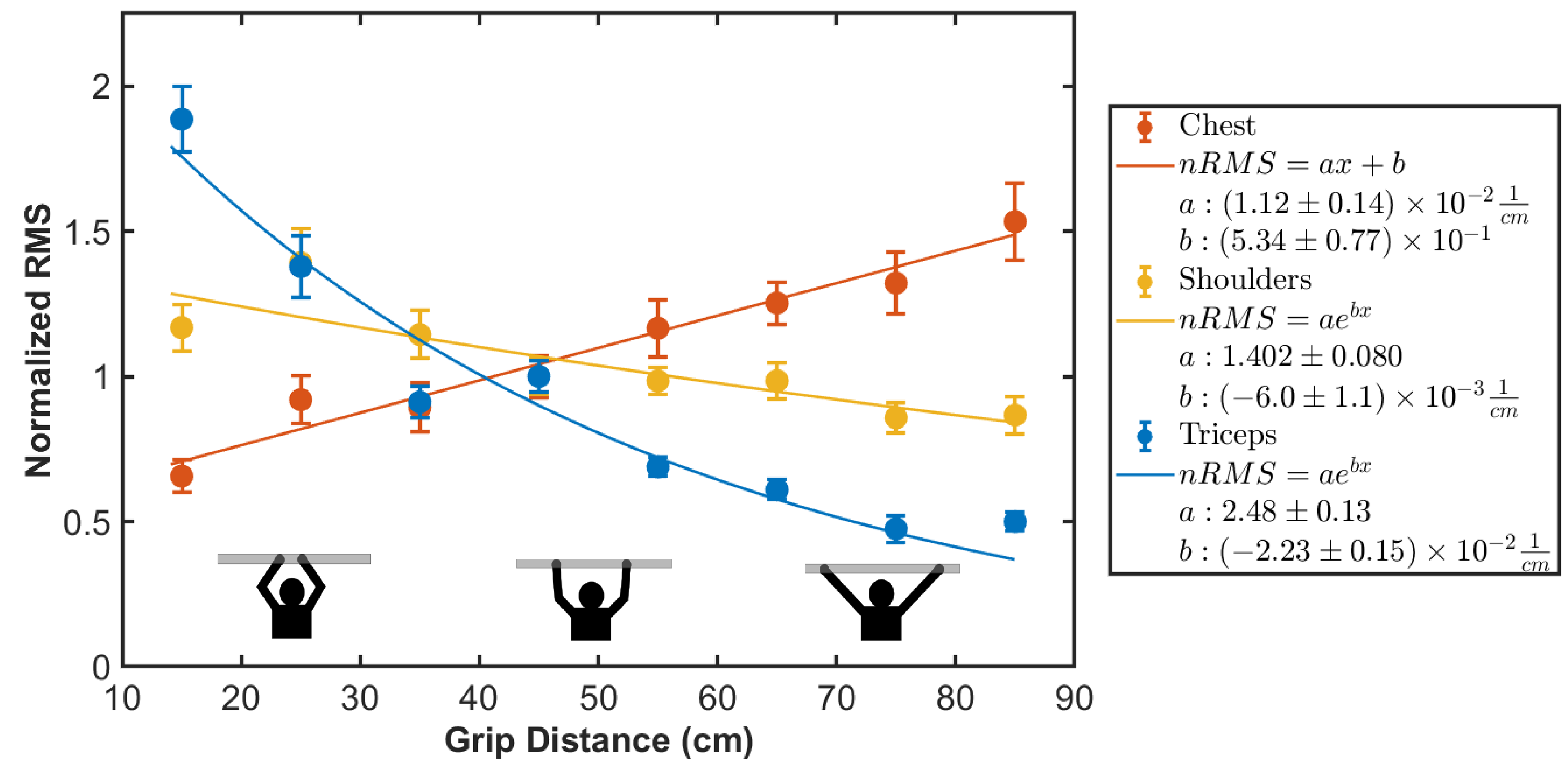
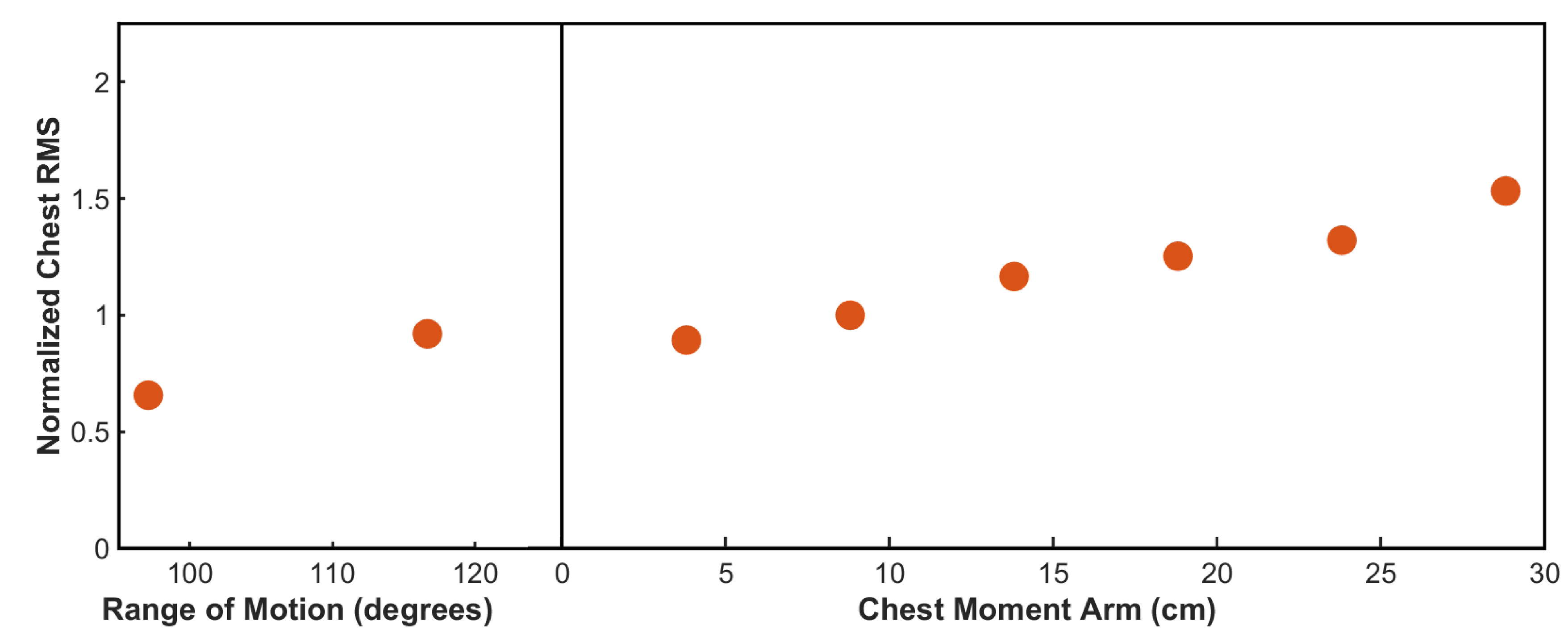
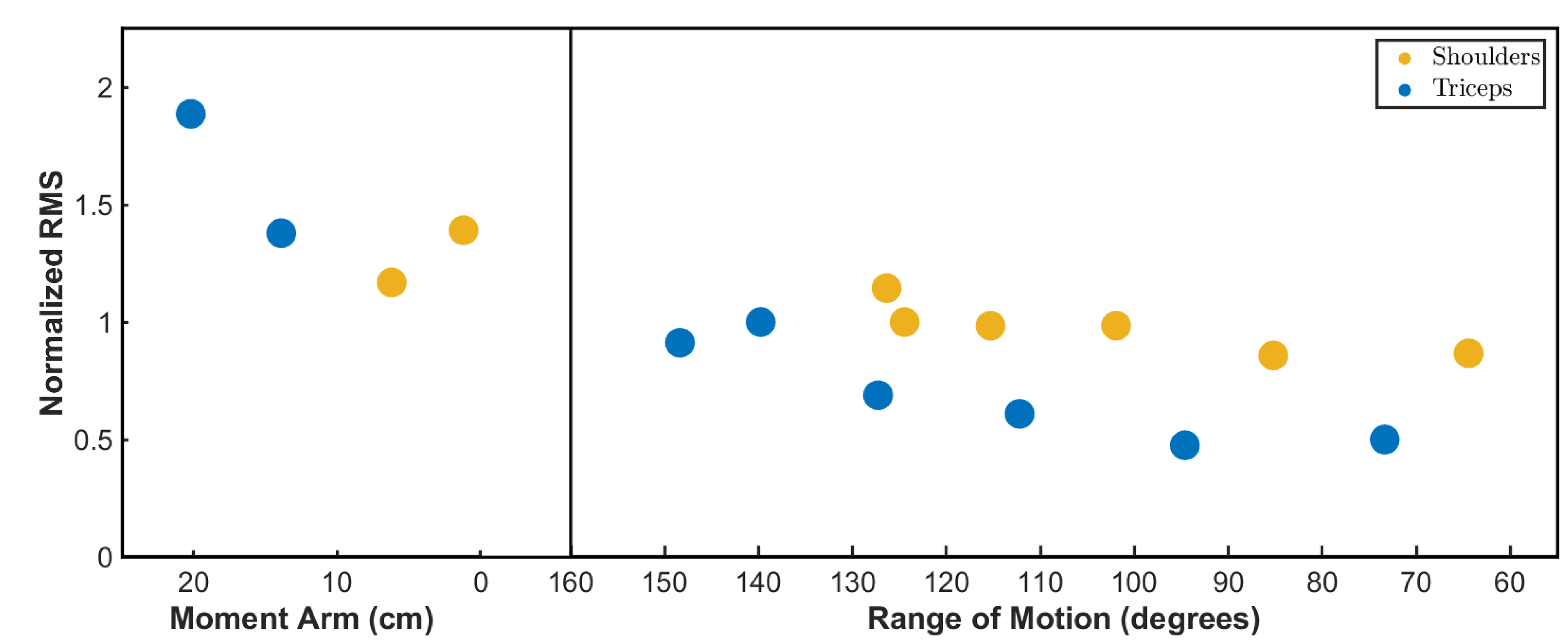
Mean RMS



Results







Conclusions

- Muscle activation changes significantly with grip distance, aligning with human biomechanical patterns.
- A grip distance near 40 cm achieves a balanced muscle activation across all three muscle groups.
- There is a trade-off between activating the chest, shoulders, and triceps. As grip distance increases, the muscle activation also increases in the chest but decreases in the shoulders and triceps.
- Wider grips target the chest, while narrower grips target the shoulders and triceps.

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References

https://imgbin.com/png/PrqyL1Ys/powerlifting-olympic-weightlifting-bench-press-png

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